On the one hand, the flexibility of a young person's human capital could argue for decreasing stocks with age. Youth has greater capacity to absorb risk—so when young you can hold more equities because you can afford the losses by absorbing them through the human capital dimension. The old person has less flexible human capital (you can't teach an old dog new tricks, supposedly), which limits the old person's ability to hold risky financial assets. On the other hand, if the flexibility of a young person's human capital is just a substitute for equities, then she will want to replace that dwindling capacity to take risk with equities as she grows older.

There are other complex and subtle interactions between labor income and capital markets. Higher wages can spur people to work harder (goofing off has a high opportunity cost). An offsetting effect is that earning a higher wage causes a person to become wealthier, and when you're really wealthy, you may really prefer to goof off.[36] Chai et al. (2011) show that the first effect dominates, so those with positive wage shocks early in life tend to work harder. These households consume more and build up relatively little financial wealth, but the savings that they do hold has a portfolio share in equities that decreases with age. The reason is that these households save through Social Security (which I comment on below) and because of high wage profiles, their Social Security benefits are higher at retirement.

3.7. SUMMARY

All else equal, the life-cycle theory predicts:

> Bond-like labor income => Reduce equities as you age.
> Stock-like labor income => Increase equities as you age.

Financial planner advice that you should reduce your equity holdings as you age (or the one hundred minus your age rule) is not appropriate for people whose labor income is positively correlated with equities. Life-cycle models predict that individuals with stock-like labor income should increase their equity holdings with age, as they wish to replace their decreasing stock-like human capital with the financial equivalent. This is what most households actually do. The need to meet health-care expenses, however, entails setting aside risk-free assets, and these risk-free positions should increase as the health-care risk rises over the life cycle.

Most financial planners and target date funds, which blindly follow versions of the one hundred minus your age rule, do not take into account how an individual's

[36] Economists label these income and substitution effects, and there is a positive relation between wages and labor supply. See Bodie, Detemple, and Rindisbacher (2009).